

IMPERIAL



**The London Register of Subsurface
CO₂ Storage
Annual Report 2025**

Overview

We report the annual amounts of CO₂ stored underground by projects globally over the period 1996 – 2024. We describe the methods used to compile this data and provide an overview of the project below. For the full Register, comprising a spreadsheet of the data with detailed information about the sources and quantification protocols used by the projects, please visit the project website at <https://imperialcollegelondon.github.io/The-London-Register-of-Subsurface-CO2-Storage/>

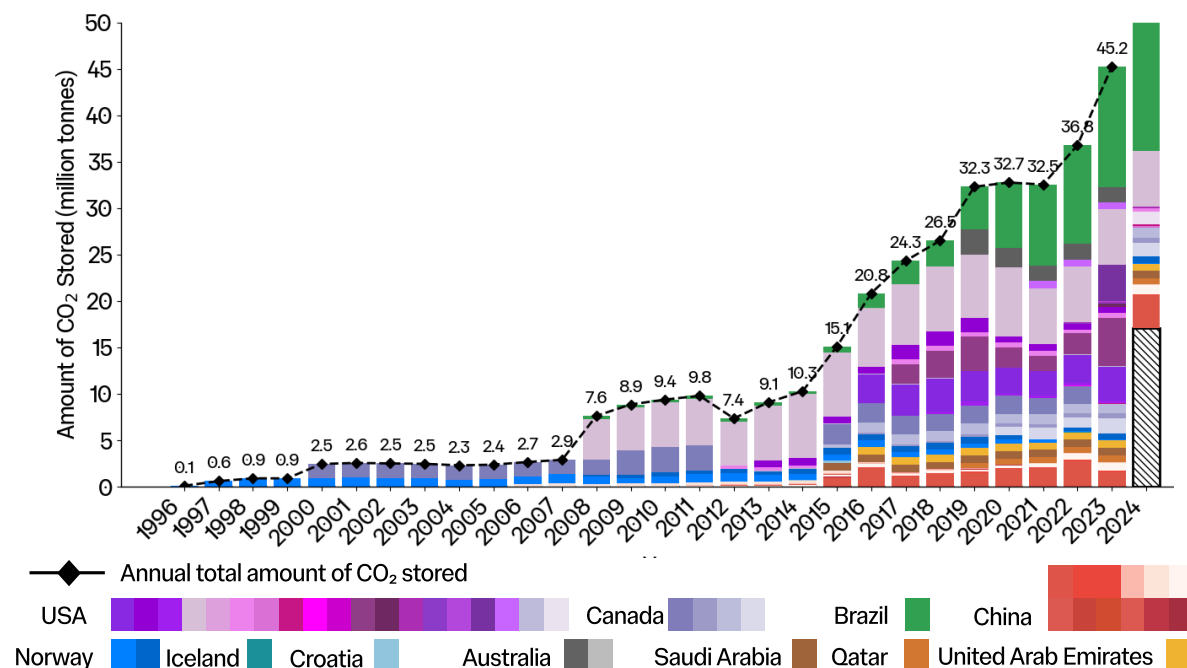


Figure 1. Annual amounts of CO₂ stored with colouring by continent, country, and individual project. See Appendix 1 for the individual project legend. The hashed grey in the bar for 2024 represents the amount of CO₂ stored in 2023 by projects where data has not yet been published

Amounts of CO₂ stored since 1996

From 0.1 million tonnes in 1996, the amount of CO₂ stored globally increased to a record 45.2 million tonnes in 2023 (Figure 1). This represents an average annual growth of 17.1% since 1996. Inclusive of 2024, a total of 384.6 million tonnes has been stored. Three distinct phases of growth can be observed

1. **Pioneering Phase (1996–2007, 3 Projects):** Annual storage amounts were less than 3 million tonnes of CO₂ per year, driven by the Sleipner project in Norway, which began in 1996 at a rate of about 0.8 million tonnes per year, and the Weyburn-Midale project in Canada, operational from 2000 at a rate of about 1.6 million tonnes per year. China's Zhongyuan oilfield began operations in 2006 with a rate of about 0.3 million tonnes per year, marking an early expansion beyond North America and Europe. The annual average growth rate of injection during this period was 21.6%, albeit with most of the growth taking place in the first year.
2. **North American Expansion (2008–2015, 18 Projects):** New projects boosted annual amounts to the range of 7–15 million tonnes of CO₂ per year. This corresponds with an annual average growth in injection rate of 6.2 % over this period. The total injection rate increase was dominated by projects in North America, and this also brought the prominence of geological storage combined with oil recovery. The Santos Basin Pre-Salt oilfield in Brazil joined the

Zhongyuan project in China in marking the initiation of a broader international deployment of CO₂ storage.

3. Global Scaling (2016–2023, 36 Projects): Large-scale deployments and new market entrants increased injection rates to 45.2 million tonnes per year with an annual average growth in injection rate of 9.8 % during this period. This phase featured a large increase in project numbers and a marked global spread, with activity ramping up in China and new large-scale projects launching in Australia, the United Arab Emirates, Saudi Arabia, and Qatar. The growing deployment in Asia, the Middle East, and South America represented an international expansion of industrial scale CO₂ storage.

Amounts of CO₂ stored in 2023

In 2023, the most recent year with complete data, there were 35 projects reporting and an aggregate reported global storage amount of 45.2 million tonnes of CO₂ (Figure 2). The total amount of stored CO₂ increased by 8.4 million tonnes from 2022, marking one of the highest single-year increases on record.

Leading the increase were projects based in the United States, contributing an additional 8.1 million tonnes. This growth was driven by both newly initiated projects and increased injection rates at existing facilities. The newly launched Seminole San Andres Unit emerged as the largest single contributor, accounting for a 3.9 million tonne increase. Other newly launched projects in the U.S.A. included the Camrick Unit, Petra Nova West Ranch, the Seminole East Field, the Blue Flint CO₂ Storage Project, and the Barnett RDC Well No. 1.

There was also considerable growth in Brazil. Expansion at Petrobras's Santos Basin Pre-Salt Oil field added 2.4 million tonnes rising from 10.6 million tonnes in 2022 to 13.0 million tonnes in 2023. As a result, CO₂ storage at the Santos Basin made up 33% of the global total for 2023.

Growth continued in Norway with an increase in injection from 0.5 million tonnes in 2022 to 0.8 million tonnes due to increases in storage rates at both Sleipner and Snøhvit.

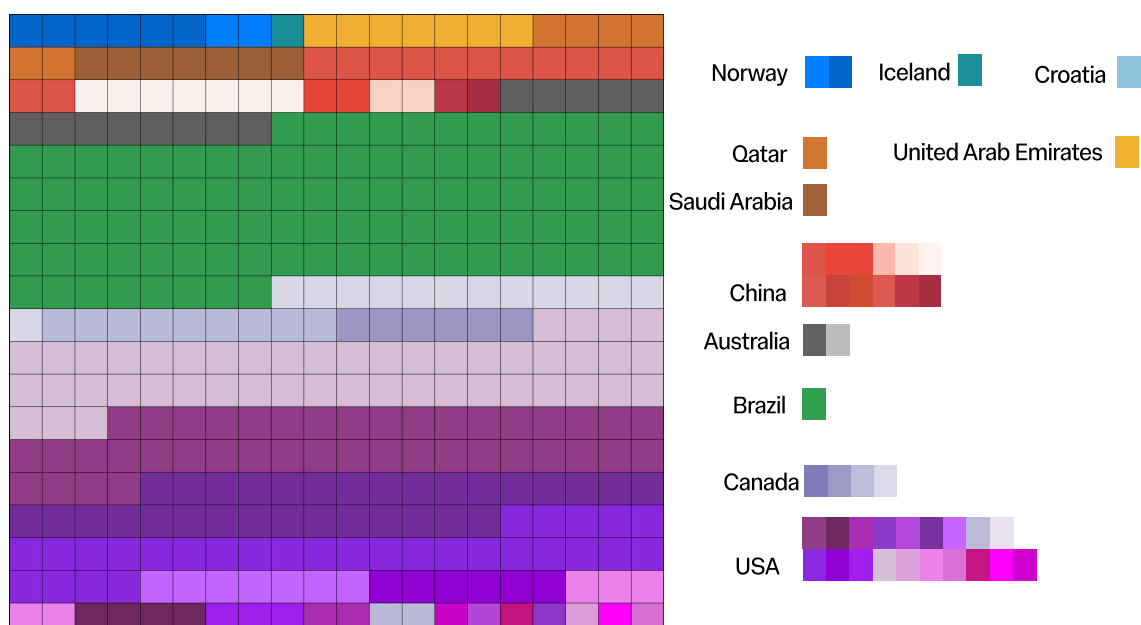


Figure 2. A waffle Plot of 2023 CO₂ Storage by country and project. The chart is a 20 × 19 grid with each block representing about 100,000 tonnes of CO₂ injected. The individual project legend is provided in Appendix 1.

However, we cannot report growth in all regions. In Canada, one major facility, Weyburn-Midale, has not yet reported injection data for 2023. This contributes to an apparent decline of 2.0 million tonnes which may change after data becomes available. Storage rates declined in China by 1.0 million tonnes, primarily due to a decline at the Jilin oilfield.

Incomplete Reporting on Amounts of CO₂ Stored in 2024

At the time of publication there were 22 projects reporting data from 2024 with an aggregated storage amount of 33.2 million tonnes. The data is currently incomplete and there is a particularly large gap in information from projects in the USA that was previously available through the EPA Greenhouse Gas Reporting Programme. This data will be back revised for the 2026 Annual Report.

Reporting for 2024 from the USA includes 6.0 million tonnes from Shute Creek and 1.4 million tonnes from the DGC Beulah Broom Creek project. Smaller amounts come from the Illinois Industrial Carbon Capture and Sequestration Project, accounting for 0.3 million tonnes, while Barnett RDC Well No. 1, the Blue Flint CO₂ Storage Project, and Red Trail Energy each storing approximately 0.2 million tonnes.

In Canada there is reporting for 3.1 million tonnes of CO₂ stored in 2024. This amount decreased slightly by 0.1 million tonnes from 2023, primarily because of a slight decline from the Alberta Carbon Trunk Line (ACTL) project.

In Brazil, the Santos Basin Pre-Salt Oil Field project continues as one of the leading contributors to the global aggregate. In 2024, 14.2 million tonnes of CO₂ were injected, a 1.2 million tonne increase from 2023. As a result, CO₂ storage at the Santos Basin project accounted for nearly half of the global reported storage for 2024.

Storage operations in China recorded a total of 4.7 million tonnes, with the Jilin Oilfield contributing 3.5 million tonnes of this figure. This marks an increase from the 2.0 million tonnes among all of China's projects reported in 2023.

The majority of Middle Eastern facilities maintained steady operations. Both Saudi Arabia and the United Arab Emirates reported unchanged levels at 0.8 million tonnes each. There was a small decrease in the storage rate in Qatar, from around 0.7 to 0.6 million tonnes.

At Sleipner and Snohvit, offshore Norway, storage rates remain the same as the previous year, with a total storage of 0.8 million tonnes in the year.

Methods

The London Register of Subsurface CO₂ Storage compiles reported amounts of CO₂ stored geologically. We summarise the methods used to compile this data here. Full details of data sources and reporting protocols can be found on the downloadable spreadsheet at the project website <https://imperialcollegelondon.github.io/The-London-Register-of-Subsurface-CO2-Storage/>.

- Data sources: The data in this report is compiled from publicly reported sources, including project-level disclosures, government databases, industry reports, environmental assessments, and data directly from project operators by response to a survey. The full register may be downloaded in spreadsheet form from the project website.
- Data processing: In many cases, multiple sources of data exist for a project in a given year, and these are included in the full database. For global injection rate calculations, and locations in this report and on our website where we provide single values for projects in a year, we have prioritised primary sources (e.g., operator responses to a survey) over secondary reporting. For some projects, only cumulative amounts of stored CO₂ is reported following several years of

activity. In these cases, we report the average value for the years of the reporting period. The specific data that are used and instances of averaging are shown in the Register spreadsheet.

- **Definitions of storage:** We accept definitions of storage as reported by projects, and these are usually governed by adherence to independent standards and regulations including those set forth by Ipieca, the Global Reporting Initiative (GRI), the Sustainability Accounting Standards Board (SASB), the Hong Kong Stock Exchange ESG Reporting Guide (HKEX-ESG), and the U.S. Environmental Protection Agency Greenhouse Gas Reporting Programme. Standards under which projects are reporting CO₂ storage are provided in the Register spreadsheet. We also provide links to some of the more commonly used standards in Appendix 2.
- **Assurance and Quality:** An independent assurance evaluation of the data has been carried out by DNV. A summary is provided in Appendix 3. The detailed outcome and recommendations can be found in a separate report issued to Imperial College London that will be available on the website.

About the London Register of Subsurface CO₂ Storage

This is a centralised register of annual rates of CO₂ stored underground from operational projects worldwide. This initiative, funded by the UK Royal Academy of Engineering and supported by a consortium of stakeholders from academia, industry, and government, obtains data directly from operators and from publicly available information, providing a standardised register of the annual amounts of stored CO₂. Please contact us with any questions or interest in the project.

This report should be cited as:

Gao, X., & Krevor, S. (2025). The London Register of Subsurface CO₂ Storage: 2025 Annual Report. Imperial College London. <https://imperialcollegelondon.github.io/The-London-Register-of-Subsurface-CO2-Storage/>

Management

Prof. Samuel Krevor – Director, Professor of Subsurface Carbon Storage, Imperial College London
s.krevor@imperial.ac.uk

Dr. Xiaowei Gao – Executive Director, Postdoctoral Researcher, Imperial College London,
x.gao@imperial.ac.uk

Dr. Anjanette Harry – Programme Manager, Imperial College London, a.harry1@imperial.ac.uk

Consortium Members

Prof. Sally Benson – Professor of Energy Science & Engineering, Stanford University

Dr Nicola Clarke – Senior Geologist, IEAGHG

Jarad Daniels – CEO, Global CCS Institute & The Global CCS Institute Team

Tim Dixon – Director and General Manager, IEAGHG

Prof. Sarah Gasda – Research Director, NORCE

Peter Golding – Program Manager, Oil and Gas Climate Initiative

Beth Hebditch – Head of UK Policy, Carbon Capture and Storage Association

Dr Samuel Jackson – Senior Research Scientist, CSIRO

Prof. Ruben Juanes – Professor of Civil and Environmental Engineering, MIT

John MacArthur – Fellow of the Royal Academy of Engineering

Dr Rachael Moore – Managing Director, CarbStrat

Erik Nickel - Chief Operating Officer and Interim CEO, Petroleum Technology Research Centre Inc

Candice Patton – Vice President, Corporate Affairs, Enhance Energy

Prof. Philip Ringrose – Professor in Energy Transition Geoscience, Norwegian University of Science and Technology

Dr Andrew Ross – Group Leader, CSIRO

Prof. Ruina Xu – Professor in Thermal Engineering, Department of Energy and Power Engineering, Tsinghua University

Appendix 1: Project Legend for Figure 1 and Figure 2

North America

Canada:

-  Alberta Carbon Trunk Line (ACTL)
-  Aquistore
-  Quest
-  Weyburn-Midale

USA:

-  30-30 Gas Plant
-  Barnett RDC Well No. 1
-  Blue Flint CO2 Storage Project
-  Campo Viejo Gas Processing Plant
-  Camrick Unit
-  Core Energy Otsego County EOR
-  DGC Beulah Broom Creek
-  Denver Unit/Wasson San Andres
-  Farnsworth Unit CO2 Flood
-  Hobbs Field
-  Illinois Industrial CCS Project
-  North Burbank Unit
-  Petra Nova West Ranch
-  Red Hills Gas Processing Plant
-  Red Trail Energy, LLC
-  Seminole East Field (SEF)
-  Seminole San Andres Unit
-  Shute Creek
-  West Seminole San Andres Unit

South America

Brazil:

-  Petrobras Santos Basin Pre-Salt Oil Field

Europe

Croatia:

-  CO2 EOR Project Croatia

Iceland:

-  Carbfix Storage

Norway:

-  Sleipner
-  Snohvit

Asia

China:

-  Caoshe Oilfield
-  Huabei Oilfield
-  Jilin Oilfield
-  Jingbian Oilfield
-  Karamay Dunhua
-  Liaohe Oilfield
-  ShengLi Oilfield
-  Shenhua Ordos
-  Xinjiang Oilfield
-  Xinjiang Ordos Basin
-  Yanchang Integrated Demonstration
-  Zhongyuan Oilfield

Japan:

-  Tomakomai CCS Demo Project

Qatar:

-  Qatar LNG

Saudi Arabia:

-  Uthmaniyah CO₂-EOR Demonstration

United Arab Emirates:

-  ADNOC AI Reyadah

Oceania

Australia:

-  CO2CRC
-  Chevron Gorgon

5 Continents | 12 Countries | 46 Projects

Appendix 2: Links to Reporting Frameworks and Standards Used by CO₂ Storage Projects in the Register

- EPA Greenhouse Gas Reporting Program Subpart RR – Geologic Sequestration of Carbon Dioxide: <https://www.epa.gov/ghgreporting/subpart-rr-geologic-sequestration-carbon-dioxide>
- GRI: <https://www.globalreporting.org/how-to-use-the-gri-standards/gri-standards-english-language/>
- HKEX-ESG standards KPIs: https://www.hkex.com.hk/-/media/hkex-market/listing/rules-and-guidance/environmental-social-and-governance/exchanges-guidance-materials-on-esg/app2_envirokpis
- Ipieca CCE-3: <https://www.ipieca.org/resources/sustainability-reporting-guidance>
- SASB Standards EM-EP: <https://sasb.ifrs.org/standards/>

Appendix 3: Data Assurance Summary appended from the following page